

Title

Assessment of polyphenol-based adjunctive therapy on renal function and biomarkers in feline chronic kidney disease

Abstract

Background: Feline chronic kidney disease commonly progresses through oxidative stress and inflammatory mechanisms. Polyphenolic compounds have demonstrated antioxidant and anti-inflammatory properties in experimental models. We propose a prospective trial to evaluate whether dietary curcumin supplementation enhances renoprotection in cats with IRIS stage 3 disease. **Methods:** Sixteen client-owned cats with confirmed stage 3 chronic kidney disease will be randomly assigned to receive either standard medical treatment alone (n=8) or standard treatment plus daily curcumin supplementation at 80 mg divided into two doses (n=8). All cats must be on stable renal diet and medication for ≥ 2 weeks prior to enrollment. Treatment will continue for 60 days with assessments at baseline, days 30, 45, and 60. **Primary outcome:** serum creatinine-independent marker concentration (symmetric dimethylarginine). **Secondary outcomes:** proteinuria, oxidative and inflammatory biomarkers, and safety/tolerability. **Inclusion:** cats ≥ 8 years with stable IRIS stage 3 disease and owner compliance capacity. **Study design:** prospective randomized controlled trial.